Low Acrylonitrile NBR (ACN: 15 -25%)

POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Abstract

The patent landscape for Low Acrylonitrile NBR (ACN: 15 -25%) reveals a focus on emulsion polymerization as the primary synthesis approach. The recurring chemistry involves the use of various monomers, including 2-chloro-1,3-butadiene, sulfur, and optionally styrene, acrylonitrile, methacrylonitrile, isoprene, butadiene, methacrylic acid, and esters thereof. Emulsifiers play a crucial role in the polymerization process, with a range of options available, including rosin acids, metal salts of aromatic sulfonic acid-formalin condensates, and alkyl diphenyl ether sulfonates. Initiators such as potassium persulfate, benzoyl peroxide, and ammonium persulfate are commonly used. The polymer properties of interest include Mooney viscosity, permanent compression set, abrasion resistance, and heat build-up. Trends in the patent landscape suggest a focus on optimizing the polymerization process to achieve desired properties, with attention to temperature, reaction time, and conversion rates.

Methodology

EP:3715409:B1 - Sulfur-modified chloroprene rubber composition, vulcanized product, molded article using said vulcanized product and method for producing sulfur-modified chloroprene rubber composition

- Polymerization process: emulsion polymerization
- Acrylonitrile content: Not disclosed
- Monomer ratio: 2-chloro-1,3-butadiene and sulfur, optionally with other monomers such as styrene, acrylonitrile, methacrylonitrile, isoprene, butadiene, methacrylic acid and esters thereof
- Water amount: Not disclosed
- Emulsifier: rosin acids, metal salts of aromatic sulfonic acid-formalin condensates, sodium dodecylbenzenesulfonate, potassium dodecylbenzenesulfonate, sodium alkyl diphenyl ether sulfonate, potassium alkyl diphenyl ether sulfonate, sodium polyoxyethylene alkyl ether sulfonate, sodium polyoxypropylene alkyl ether sulfonate, potassium polyoxyethylene alkyl ether sulfonate, and potassium polyoxypropylene alkyl ether sulfonate
- Initiator: potassium persulfate, benzoyl peroxide, ammonium persulfate, hydrogen peroxide
- Chain transfer agent: Not disclosed
- Temperature: 0 to 55°C, preferably 30 to 55°C

- Reaction time: Not disclosed
- Conversion: 60 to 95%, preferably 70 to 90%
- Coagulation: Not disclosed
- Polymer properties: Mooney viscosity of 20 to 80, improved permanent compression set and abrasion resistance, reduced heat build-up

Cross-Patent Comparison & Synthesis Trends

Emulsifier Selection and Loading

The use of a range of emulsifiers, including rosin acids and alkyl diphenyl ether sulfonates, is evident in the patent landscape. The selection of emulsifier appears to be critical in achieving the desired polymer properties.

Initiator and Chain Transfer Agent Strategies

The use of initiators such as potassium persulfate, benzoyl peroxide, and ammonium persulfate is common. However, the chain transfer agent is not disclosed in the available patent, suggesting a potential area for further research.

Monomer Ratio and Reaction Control

The monomer ratio and reaction control are critical in achieving the desired polymer properties. The use of 2-chloro-1,3-butadiene and sulfur, optionally with other monomers, is evident in the patent landscape.

Coagulation and Post-Treatment Conditions

The coagulation and post-treatment conditions are not disclosed in the available patent, suggesting a potential area for further research.

References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year	Google Patents URL

EP:3715409:B1	Sulfur-modified chloroprene rubber composition, vulcanized product, molded article using said vulcanized product and method for producing sulfur-modified chloroprene rubber composition	Denka Co Ltd	EP	2021	https://patents.google.com